

Linux Cheat-Sheet — Kali Linux

Task 1 | Cybersecurity & Ethical Hacking Internship | ApexPlanet

A quick-reference guide for essential Linux commands used in penetration testing.

File System Navigation

```
pwd # Print current working directory
ls # List files in current directory
ls -la # List all files with permissions (hidden too)
ls -lh # Human-readable file sizes
cd /path/to/dir # Change directory
cd ~ # Go to home directory
cd .. # Go up one directory
cd - # Go to previous directory
```

File & Directory Operations

```
touch file.txt # Create an empty file
mkdir foldername # Create a new directory
mkdir -p a/b/c # Create nested directories
cp file.txt /dest/ # Copy file to destination
cp -r folder/ /dest/ # Copy folder recursively
mv file.txt newname # Rename or move a file
rm file.txt # Delete a file
rm -rf foldername # Delete folder and all contents
cat file.txt # Display file contents
less file.txt # Scroll through file (q to quit)
head -n 20 file.txt # Show first 20 lines
tail -n 20 file.txt # Show last 20 lines
nano file.txt # Edit file with nano
```

File Permissions

```
# Permission format: [type][owner][group][others]
# -rwxr-xr-- = file, owner=rwx, group=r-x, others=r--
chmod 755 file.sh # rwxr-xr-x
chmod 644 file.txt # rw-r--r--
chmod 777 file.sh # rwxrwxrwx (use carefully)
chmod +x script.sh # Add execute permission
chown user file.txt # Change file owner
chown user:group file.txt # Change owner and group
chown -R user:group folder/ # Recursive change
# Values: 4=read 2=write 1=execute
# 7=rwx | 6=rw- | 5=r-x | 4=r-- | 0=---
```

Package Management (Debian/Kali)

```
apt update # Update package list
apt upgrade # Upgrade all packages
apt install nmap # Install a package
apt install nmap wireshark # Install multiple packages
apt remove nmap # Remove a package
apt purge nmap # Remove package + config files
apt autoremove # Remove unused dependencies
apt search keyword # Search for a package
dpkg -i package.deb # Install a .deb file manually
dpkg -l | grep nmap # Check if package is installed
```

Networking Commands

```
ifconfig # Show network interfaces (legacy)
ip a # Show all interfaces (modern)
ip addr show eth0 # Show specific interface
ping 192.168.56.1 # Send ICMP packets to host
ping -c 4 google.com # Ping 4 times then stop
netstat -tulnp # Show open ports and services
ss -tulnp # Modern alternative to netstat
traceroute 8.8.8.8 # Trace route to host
route -n # Show routing table
arp -a # Show ARP table
```

Nmap — Network Scanning

```
# Basic Scans
nmap 192.168.56.1 # Basic scan (top 1000 ports)
nmap 192.168.56.0/24 # Scan entire subnet
nmap -p 80,443 192.168.56.1 # Scan specific ports
nmap -p- 192.168.56.1 # Scan ALL 65535 ports

# Scan Types
nmap -sS 192.168.56.1 # SYN/Stealth scan (root)
nmap -sT 192.168.56.1 # TCP Connect scan
nmap -sU 192.168.56.1 # UDP scan
nmap -sP 192.168.56.0/24 # Ping sweep

# Service & OS Detection
nmap -sV 192.168.56.1 # Detect service versions
nmap -O 192.168.56.1 # OS detection
nmap -A 192.168.56.1 # Aggressive: all detection

# Output
nmap -oN scan.txt 192.168.56.1 # Save to text file
nmap -oX scan.xml 192.168.56.1 # Save to XML
nmap -v 192.168.56.1 # Verbose mode
```

Wireshark / tcpdump — Packet Capture

```
tcpdump -i eth0 # Capture on eth0
tcpdump -i eth0 -w capture.pcap # Save to file
tcpdump -r capture.pcap # Read saved capture
tcpdump -i eth0 port 80 # HTTP traffic only
tcpdump -i eth0 host 192.168.56.1 # Traffic to/from host
tcpdump -i eth0 tcp # TCP packets only

# Wireshark Display Filters (GUI)
# http -> HTTP traffic only
# tcp.port == 21 -> FTP traffic
# ip.addr == 192.168.56.1 -> Traffic to/from IP
# tcp.flags.syn == 1 -> SYN packets
# ftp contains "PASS" -> FTP passwords
```

Useful Pentesting Commands

```
# Netcat
nc -lvp 4444 # Listen on port 4444
nc 192.168.56.1 4444 # Connect to host
nc -z 192.168.56.1 20-100 # Port scan range

# SSH
ssh user@192.168.56.1 # Connect via SSH
ssh -p 2222 user@host # SSH on custom port

# cURL
curl http://192.168.56.1 # GET request
curl -I http://192.168.56.1 # Headers only

# OpenSSL
openssl s_client -connect host:443 # Inspect SSL cert
echo "text" | openssl dgst -sha256 # SHA256 hash
openssl enc -aes-256-cbc -in f -out f.enc # Encrypt
```

User & Process Management

```
whoami # Show current user
id # Show user ID and groups
sudo command # Run command as root
sudo -i # Open root shell

# Processes
ps aux # List all running processes
ps aux | grep nmap # Find specific process
top # Live process viewer
kill PID # Kill process by ID
kill -9 PID # Force kill
killall nmap # Kill all named processes

# Users
cat /etc/passwd # List all users
cat /etc/shadow # Password hashes (root only)
```

Search & Text Processing

```
grep "pattern" file.txt # Search for pattern
grep -r "pattern" /dir/ # Recursive search
grep -i "pattern" file.txt # Case-insensitive
grep -n "pattern" file.txt # Show line numbers
find / -name "*.txt" # Find all .txt files
find /home -user aryan # Files owned by user
find / -perm /4000 # Find SUID files
cat file.txt | sort # Sort lines
cat file.txt | wc -l # Count lines
cut -d: -f1 /etc/passwd # Extract usernames
sed 's/old/new/g' file.txt # Find and replace
```

Shell Tips & Tricks

```
!! # Repeat last command
Ctrl + C # Cancel running command
Ctrl + Z # Suspend process
Ctrl + R # Reverse search history
Ctrl + L # Clear screen
Tab # Auto-complete

# Redirect output
command > file.txt # Write output (overwrite)
command >> file.txt # Append output
command 2> errors.txt # Redirect errors
command 2>&1 | tee out.txt # Save stdout & stderr

# Background jobs
command & # Run in background
jobs # List background jobs
fg # Bring to foreground
```

Quick Reference Card

Task	Command
Current directory?	pwd
List files	ls -la
Who am I?	whoami
My IP address	ip a / ifconfig
Scan a host	nmap -sV <ip>
Capture packets	tcpdump -i eth0
Install a tool	apt install <tool>
Make script executable	chmod +x script.sh
Connect via SSH	ssh user@<ip>
Search in file	grep "term" file
Find a file	find / -name file.txt
Running processes	ps aux
Kill a process	kill -9 <PID>
Check open ports	netstat -tulnp
Read a file	cat file.txt

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